

eHomes

Electronic homes, for both control and security

Have you ever landed in mid-winter, wishing you could telephone your flat and have the heating turned on and bath water warming up? Intrigued by the possibility, Edward Henning went along to visit a couple of eHomes projects from Designarch, and meet the architect responsible, J. K. Jain.

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J. K. Jain uses the term eHome – which he has trademarked – to mean much more than just electronic-home. It is that aspect that is of most interest to us, but we should first describe the full range of meanings. There are four specific intentions in the use of this term:

Electronically operated: both for control of systems in the home and for security.

Eco-friendly: energy conservation, better hygiene and reduced environmental impact.

Earthquake resistance: taking into consideration the soil type and foundation of a building and the materials and design.

Entertainment: the facilities available with the home.

We'll take these in reverse order, in increasing interest to us here.

Entertainment

The eHomes projects consist of high-rise block in two locations, both near Delhi – Vaishali and Greater Noida. The entertainment aspect mainly involves communal facilities that are available to all the apartments in the blocks. These include such things as health club and swimming pool, a small "home" theatre (couple

of dozen seats), communal library, billiard room, steam room and sauna – with more planned. A key feature is that a resident can check availability of these services from the electronic control system in the home and also make bookings. This is being extended to home delivery from the local mall, where one exists, and also certain local shops.

Earthquake resistance

This is a topic in which J. K. Jain has taken a particular interest over many years, and in which he is considered something of an expert. He explained that, like most developers, his group extracts ground water for mixing concrete, but unlike many, they purify the water before use. This removes chemicals that would be likely to react with the steel reinforcing in the concrete and cause deterioration over time, with the concrete separating from the steel, thereby losing strength.



The frame structure of these buildings is also strengthened by the addition of extra rings at junctions, and fly ash (a residue from burning coal) is used to produce a more rich and dense, hence stronger, form of concrete; one that also strengthens over time as it dries out and is generally more durable. Less water is needed to produce concrete using fly ash, which is also a recycled material, otherwise a waste product, and its use thereby contributes to the eco-friendliness of the buildings.

Eco-friendly

There are several aspects of these buildings that contribute to a more eco-friendly approach. The most obvious is energy conservation, and there are many ways in which this is achieved. One is the use of foam concrete blocks – partly composed of the just-mentioned fly ash. These provide generally better thermal insulation. Also providing extra insulation are uPVC windows. everybody has heard